RaceDay - Part 1 Planning Documents

Folder contents

- RaceDay_ERD.pdf - Entity Relationship Diagram.
- RaceDay_ERD.png - PNG version of the ERD.
- RaceDay_API_Endpoint_Plan.md - REST API endpoint plan.
- RaceDay_Database.sql - SQL Server schema, constraints and seed data.

ERD and SQL consistency

The ERD contains seven entities: Users, Organisers, Participants, Events, Categories, Enrolments and Results. The SQL script implements these same seven entities and the same primary-key/foreign-key relationships.

Main relationships

- Users 1-to-1 Organisers
- Users 1-to-1 Participants
- Organisers 1-to-many Events
- Events 1-to-many Categories
- Participants 1-to-many Enrolments
- Categories 1-to-many Enrolments
- Enrolments 1-to-0..1 Results

The many-to-many business relationship between Participants and Categories is resolved through Enrolments.

Testing

Open RaceDay_Database.sql in SQL Server Management Studio and execute it on a clean SQL Server instance. The script creates RaceDayDB, creates all tables and constraints, inserts sample data, and runs verification SELECT statements.